

Project TITLE

CREDIT CARD APPROVAL PREDICTION

Emotion Detection & Learning Support Engine Using Machine Learning

Team

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Abstract

The “Emotion Detection & Learning Support Engine Using Machine Learning” project is designed to identify and analyze the emotional state of students and provide suitable learning support based on their emotions. In modern education systems, understanding student emotions plays an important role in improving learning performance, engagement, and mental well-being.

This project uses Machine Learning techniques to detect emotions such as happy, sad, angry, stressed, or neutral from user inputs such as facial expressions, text, or voice data. After detecting the emotion, the system provides personalized learning support and recommendations to help students learn effectively.

The system includes data collection, preprocessing, feature extraction, model training, emotion prediction, and recommendation generation. Various Machine Learning algorithms are used to improve prediction accuracy and system performance.

The main objective of this project is to create an intelligent educational support system that enhances the learning experience by understanding emotional behavior. The project demonstrates the practical application of Artificial Intelligence and Machine Learning in the field of education and student support systems.

FEATURES

- User-friendly web interface for easy interaction
- Predicts credit card approval status instantly
- Uses Machine Learning algorithms for accurate prediction
- Performs data preprocessing and validation
- Analyzes applicant financial and personal details
- Reduces manual verification process
- Fast and automated decision-making system
- Improves prediction accuracy using trained datasets
- Displays approval or rejection results clearly
- Supports real-time prediction through deployment
- Helps reduce financial risk for banks and institutions
- Scalable and efficient prediction system

Tech Stack

Frontend:

- HTML
- CSS
- JavaScript

Backend:

- Python
- Flask

Machine Learning:

- Scikit-learn
- Pandas
- NumPy

Visualization & Development Tools:

- Matplotlib
- Jupyter Notebook

Deployment Platform:

- Render

Database:

- CSV Dataset

INSTALLATION STEPS

1. Install Python on the system.
2. Install required Python libraries using the following command:
`pip install flask pandas numpy scikit-learn matplotlib`
3. Download or clone the project source code.
4. Open the project folder in VS Code or any Python IDE.
5. Run the application using the command:
`python app.py`
6. Open the browser and enter the local host URL:
<http://127.0.0.1:5000/>
7. Enter applicant details in the prediction form.
8. Click on the Predict button to view the credit card approval result.
9. For online access, deploy the project using the Render platform.

USAGE INSTRUCTIONS

1. Open the deployed application using the provided deployment link.
2. The home page of the Credit Card Approval Prediction system will be displayed.
3. Enter the required applicant details such as income, age, employment status, credit history, and other financial information.
4. Verify that all input fields are filled correctly.
5. Click on the “Predict” button to submit the application details.
6. The Machine Learning model processes the entered data and analyzes the applicant information.
7. The system displays the prediction result as either Approved or Rejected.
8. Users can test multiple applicant details to analyze different prediction outcomes.
9. The application can be accessed through any web browser with an internet connection.

Future Scope

- Improve emotion classification accuracy using larger datasets.
- Integrate real-time transformer inference.
- Support multilingual emotion detection.
- Provide personalized learning recommendations.
- Deploy the application as a cloud-based web service.

Epics

- Epic 1 – ERD diagram , Technology, Workflow, documentation (Jayavarapu Sai Meghana)
- Epic 2 – Importing the libraries, read the dataset, univariate& multivariate analysis, descriptive analysis(Himavardhan Abbina)
- Epic 3 – Drop duplicate features, handling missing values, data cleaning && merging, feature engineering, handling categorical values(Karri Sri Rama Krishna)
- Epic 4 – Logistic Regression model, Random forest ,Decision tree model (Veerendra Mohan Sripadam)
- Epic 5 – Building HTML pages, Build the python script, Run the application(Bandaru Radha Krishna)

Epic 1 - **PROJECT PLANNING AND SYSTEM DESIGN OVERVIEW** (Jayavarapu Sai Meghana)

Overview

This phase focuses on the initial planning and design of the Credit Card Approval Prediction system. The project requirements, workflow structure, technology stack, ERD diagram, and documentation process were prepared during this stage. The overall system architecture and development approach were finalized to ensure smooth project implementation.

OBJECTIVE

- To understand the project requirements
- To design the workflow of the system
- To prepare the ERD and system architecture
- To identify suitable technologies for development
- To create proper project documentation and planning

Tasks Completed

- Created ERD diagram
- Designed project workflow
- Selected technologies and tools
- Prepared project documentation
- Planned system architecture

Technologies Used

- Python
- Flask
- HTML/CSS
- Draw.io / Lucidchart
- VS Code

Development Environment

- Windows Operating System
- Visual Studio Code
- Jupyter Notebook
- Google Chrome Browser

EPIC 2 – DATA ANALYSIS AND DATASET UNDERSTANDING(Himavardhan Abbina)

OVERVIEW

This phase focuses on importing the required libraries, loading the dataset, and analyzing the data. Various analytical methods such as univariate analysis, multivariate analysis, and descriptive statistics were performed to understand the dataset structure and identify important patterns for prediction.

OBJECTIVE

- To import required Machine Learning libraries
- To load and understand the dataset
- To perform data analysis and visualization
- To identify relationships between features
- To understand statistical information of the dataset

Tasks Completed

- Imported required libraries
- Loaded dataset using Pandas
- Performed univariate analysis
- Performed multivariate analysis
- Conducted descriptive statistical analysis
- Visualized important data patterns

Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Jupyter Notebook

Development Environment

- Jupyter Notebook
- VS Code
- Python Environment

EPIC 3 – DATA PREPROCESSING AND FEATURE ENGINEERING

OVERVIEW

(Karri Sri Rama Krishna)

This phase focuses on cleaning and preprocessing the dataset to improve data quality and model performance. Duplicate records, missing values, and inconsistent data were handled carefully. Feature engineering and categorical value transformation techniques were also applied to prepare the dataset for Machine Learning model training.

OBJECTIVE

- To clean and preprocess the dataset
- To remove duplicate and unwanted features
- To handle missing values effectively
- To perform feature engineering techniques
- To convert categorical values into numerical format

Tasks Completed

- Removed duplicate features
- Handled missing values
- Performed data cleaning
- Merged required data fields
- Applied feature engineering
- Encoded categorical values
- Prepared final dataset for training
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Technologies Used

- Python
- Pandas
- NumPy
- Scikit-learn

Development Environment

- Jupyter Notebook
- VS Code
- Python Environment

EPIC 4 – MACHINE LEARNING MODEL DEVELOPMENT

(Veerendra Mohan Sripadam)

OVERVIEW

This phase focuses on building and training different Machine Learning models for credit card approval prediction. Multiple algorithms such as Logistic Regression, Random Forest, and Decision Tree were implemented and evaluated to identify the best-performing prediction model.

OBJECTIVE

- To train Machine Learning models
- To compare model performance and accuracy
- To improve prediction efficiency
- To select the best prediction model
- To evaluate system performance

Tasks Completed

- Implemented Logistic Regression model
- Implemented Random Forest model
- Implemented Decision Tree model
- Trained and tested models
- Compared model accuracy
- Evaluated prediction results

Technologies Used

- Python
- Scikit-learn
- Pandas
- NumPy
- Matplotlib

Development Environment

- Jupyter Notebook
- VS Code
- Python Environment

EPIC 5 – APPLICATION DEVELOPMENT AND DEPLOYMENT

(Bandaru Radha Krishna)

OVERVIEW

This phase focuses on developing the user interface and integrating the Machine Learning model into the web application. HTML pages and Python backend scripts were created to provide prediction functionality. The application was tested and executed successfully for real-time credit card approval prediction.

OBJECTIVE

- To develop the frontend user interface
- To integrate Machine Learning models with the backend
- To build prediction functionality
- To execute and test the application
- To deploy the application for real-time usage

Tasks Completed

- Designed HTML pages
- Developed Python backend scripts
- Integrated trained ML model
- Connected frontend and backend
- Tested application functionality
- Ran the application successfully
- Deployed project on Render platform

Technologies Used

- HTML
- CSS
- JavaScript
- Python
- Flask
- Render

Development Environment

- VS Code
- Python Environment
- Web Browser
- Render Cloud Platform